

Matrices

4.10 Similarity Transformation

Eigen basis $\rightarrow$ Diagonal matrix

$A \rightarrow n \times n$ (Given), $\hat{A} \rightarrow n \times n$ (Similar matrix)

$\hat{A} = P^{-1}AP \Rightarrow \hat{A}$ is similar to A

$\hat{A} = P^{-1}AP$: This transformation is known as similarity transformation.

- It preserves the eigenvalues of A .
- If x is an eigenvector of A , then $y = P^{-1}x$ is an eigenvector of $\hat{A}$ corresponding to the same eigenvalue.
- A matrix P can be reflected such that the similarity transformation results into a D . (diagonal matrix)